

<https://www.simoescostalab.org/>

Introduce topic;
State *knowledge gap*

Overview of Research Projects

Project: Investigating Pathogen Infection in the Intestine, Heriot-Watt, London

Specific Aims

Microscopy has emerged as one of the most powerful and informative ways to analyse cell-based high-throughput screening (HTS) samples in experiments designed to uncover novel drugs and drug targets. However, many diseases and biological pathways can be better studied in whole animals, particularly diseases that involve organ systems and multifactorial interactions, such as metabolism and infection. The worm *Caenorhabditis elegans* is a well-established and effective model organism that can be relatively prepared and imaged, for existing image analysis methods are sufficient for most assays.

We propose to develop algorithms for the analysis of high-throughput *C. elegans* images, validating them in three specific experiments to identify chemicals to curb human infections and genetic regulators of host response to pathogens and fat metabolism. Novel computational tools for automated image analysis of *C. elegans* assays will make whole animal screening possible for a variety of biological questions not amenable to cell-based assays. Building on our expertise in developing image processing and machine learning algorithms for high-throughput screening, and on our established collaborations with leaders in *C. elegans* research, we will:

Aim 1: Develop algorithms for *C. elegans* viability assays to identify modulators of pathogen infection
Challenge: To identify individual worms in thousands of two-dimensional brightfield images of worm populations infected by *Mycobacterium*, and measure viability based on worm body shape. The worms are very similar to dead worms and are straight.
Approach: We will develop algorithms that use a probabilistic shape model of *C. elegans* learned from examples, enabling segmentation and body shape measurements even when worms touch or cross.
Impact: These algorithms will quantify a wide range of phenotypic descriptors detectable in individual worms, including body morphology as well as subtle variations in reporter signal levels.

Aim 2: Develop algorithms for *C. elegans* lipid assays to identify genes that regulate fat metabolism
Challenge: To detect worms versus background, despite artifacts from sample preparation, and detect subtle phenotypes of worm populations.
Approach: We will improve self-edge detection, illumination correction, and detection of artifacts (e.g. bubbles and aggregates of bacteria) and enable image segmentation in highly variable image backgrounds using level set segmentation. We will also design feature descriptors that can capture worm population phenotypes.
Impact: These algorithms will provide information for a variety of phenotypes in worm populations. They will also improve data quality in other assays, such as those in Aims 1 and 3.

Aim 3: Develop algorithms for gene expression pattern assays to identify regulators of the response of the *C. elegans* host to *Staphylococcus aureus* infection
Challenge: To map each worm to a reference and quantify changes in fluorescence localization patterns. This work will provide the *C. elegans* community with (a) a versatile, modular, open-source toolbox of algorithms readily usable by biologists to quantify a wide range of important high-throughput whole-organism assays, (b) a new framework for extracting morphological features from *C. elegans* populations for quantitative analysis of this organism, and (c) the capability to discover disease-related pathways, chemical probes, and drug targets in high-throughput screens relevant to a variety of diseases.

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Describe specific aims
(generally 2 or 3)

Each is a testable objective that
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Deliverables,
broad impacts

Overview of Research Projects

Research Projects

will be presented on Friday (7/3) by
6 groups of 5 students

- Presentations will be 30mins + 10mins of Q&A;
- Projects will be evaluated as a grant application;
- Any topic is OK as long as it is centered in **genomics** and uses techniques discussed in class.

Overview of Research Projects

Title / big question

One slide. What is the biological problem?

Background and significance

3–5 slides. What is known? Why does it matter?

Knowledge gap

1 slide. What do we still not understand?

Central hypothesis / overall objective

1 slide. What do you want to accomplish

Specific Aims

Ideally 2-3 aims, 2-3 slides per aim. For each aim:

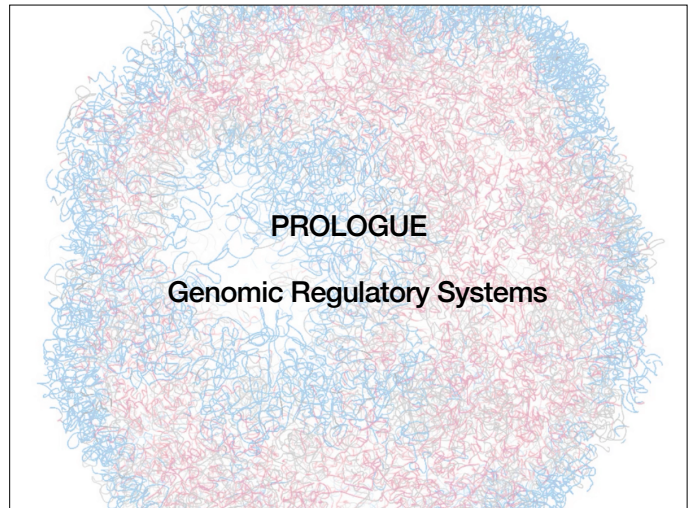
- What question does this aim answer?
- What techniques will be used?
- What are the key controls?
- What result would support or refute the hypothesis?
- What are possible pitfalls/alternatives?

Deliverables and impact

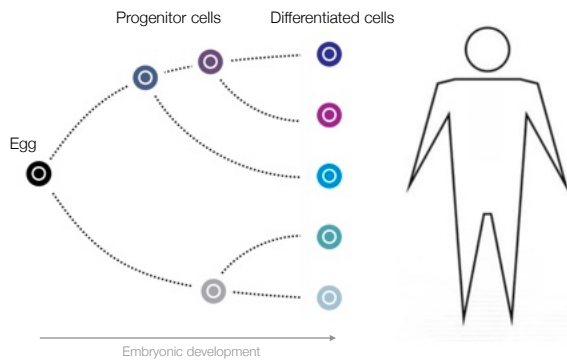
1–2 slides. What will the field know after this project? Why is it important?

PROLOGUE

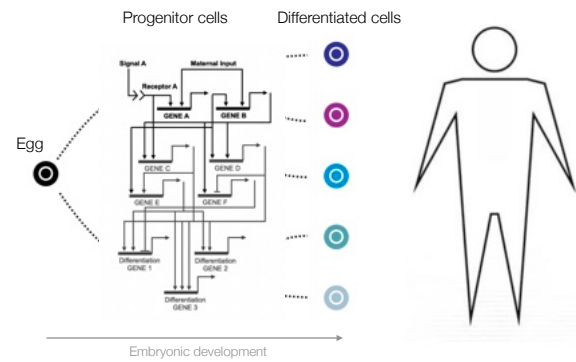
Genomic Regulatory Systems



Cell identity is governed by gene regulatory networks (GRNs)



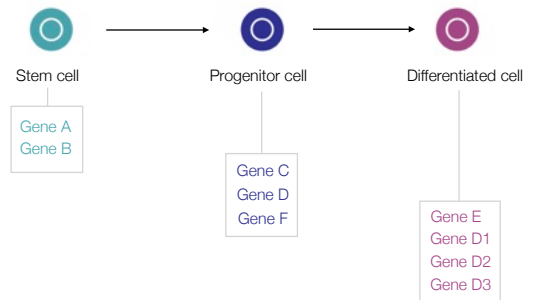
Cell identity is governed by gene regulatory networks (GRNs)



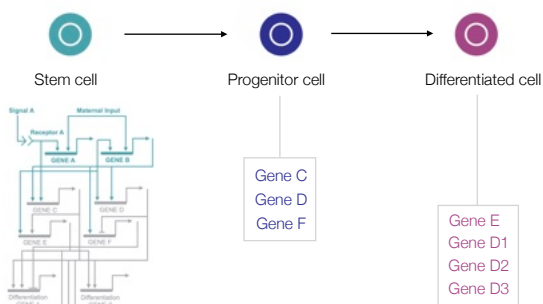
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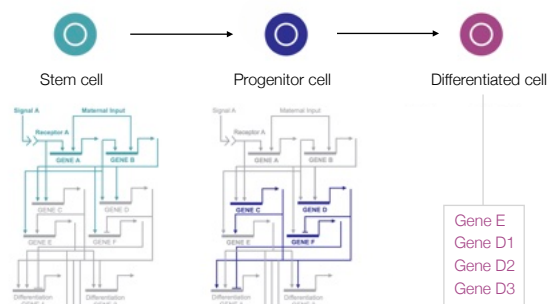
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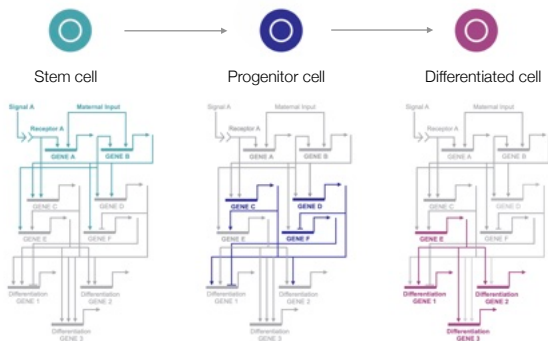
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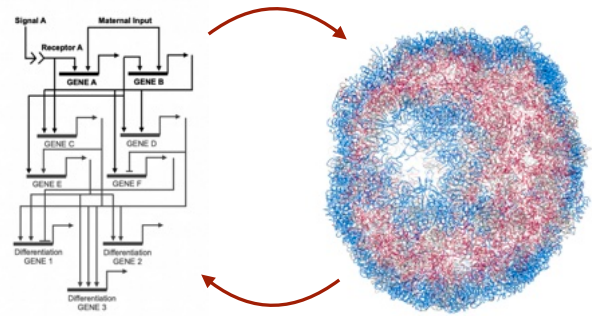
Cell identity is governed by gene regulatory networks (GRNs)



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The genome is the physical substrate that processes regulatory information



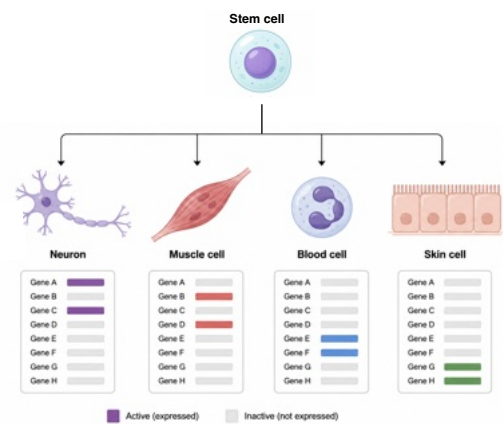
GRNs are not abstract circuits: they operate through enhancers, promoters, chromatin accessibility, and 3D genome folding

PART I:

Differential Gene Expression



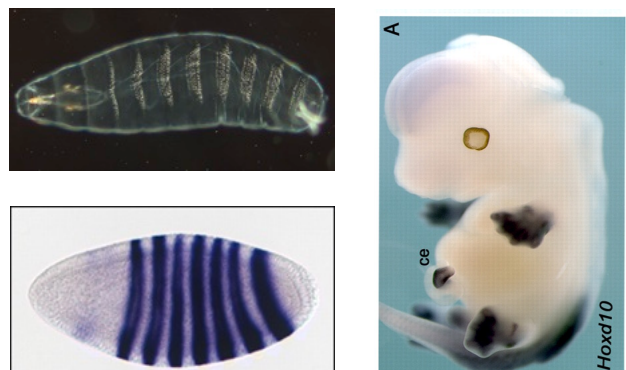
What makes cells different from each other?



Differential gene expression is a central part of embryonic development



Differential gene expression is a central part of embryonic development



Left side mutations:

ASD: NKX2-5	VSD: NKX2-5	DORV: TA: NKX2-5
GA434	GA434	TA: NKX2-5
TBAG2	TRK3	PTA: TBAG2
SRFBP1	TRK3	TGA: NKX2-5
TRK3	TRK3	TOF: NKX2-5

Right side mutations:

AVSD: NKX2-5	AS and AC: NKX2-5
SOX9	HLHS: NKX2-5
RAFI	PS: PITX1
CHD1	NKX2-5
Ebstein, TA: NKX2-5	

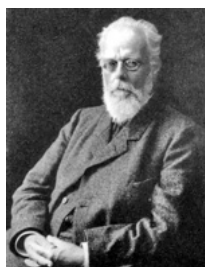
The diagram illustrates the evolution of a cell population through various phenotypic states and transitions. The process begins with a single cell (grey) undergoing "Mutation-driven proliferation" to form a population of "Differentiated or proliferative phenotype" cells (blue). This population then undergoes "Proliferation" to form a larger population of blue cells. From this population, two pathways emerge: one leading to "Transcriptional plasticity" (a mix of blue and yellow cells) and another leading to "Intermediate phenotype" (a mix of blue and green cells). The "Transcriptional plasticity" population can further evolve into a "Dedifferentiated phenotype" (yellow cells) or undergo "Cooperative invasion" (a mix of blue, yellow, and green cells). The "Intermediate phenotype" population can undergo "Cooperative invasion" or "Heterogeneous dissemination" (a mix of blue, yellow, and green cells). The "Dedifferentiated phenotype" population can undergo "Seeding" (a mix of blue and yellow cells) or "Proliferation or transcriptional plasticity" (a mix of blue and yellow cells). The "Cooperative invasion" population can undergo "Cooperative dissemination" (a mix of blue, yellow, and green cells) or "Proliferation or transcriptional plasticity" (a mix of blue and yellow cells). The "Heterogeneous dissemination" population can undergo "Seeding" (a mix of blue and yellow cells) or "Proliferation or transcriptional plasticity" (a mix of blue and yellow cells).

Stem cell

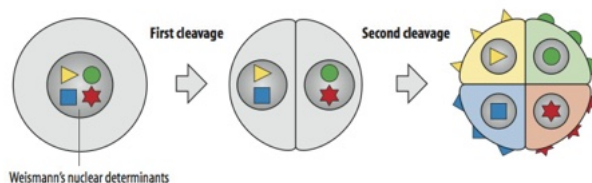
The stem cell differentiates into four specialized cell types:

- Neuron**: Morphology of a neuron. Gene expression: Gene A (Active), Gene B (Inactive), Gene C (Active), Gene D (Inactive), Gene E (Inactive), Gene F (Inactive), Gene G (Inactive), Gene H (Inactive).
- Muscle cell**: Morphology of a muscle cell. Gene expression: Gene A (Inactive), Gene B (Active), Gene C (Inactive), Gene D (Active), Gene E (Inactive), Gene F (Inactive), Gene G (Inactive), Gene H (Inactive).
- Blood cell**: Morphology of a blood cell. Gene expression: Gene A (Inactive), Gene B (Inactive), Gene C (Inactive), Gene D (Inactive), Gene E (Inactive), Gene F (Active), Gene G (Inactive), Gene H (Inactive).
- Skin cell**: Morphology of a skin cell. Gene expression: Gene A (Inactive), Gene B (Inactive), Gene C (Inactive), Gene D (Inactive), Gene E (Inactive), Gene F (Inactive), Gene G (Active), Gene H (Active).

Legend: Active (expressed) (Purple bar), Inactive (not expressed) (Grey bar).

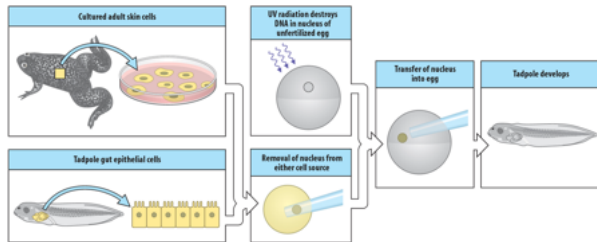


Cell fate is determined by asymmetric inheritance of the germ plasm



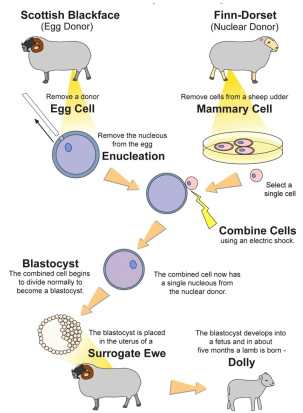
Cloning and nuclear equivalence

Sir John Gurdon
Nobel Prize 2012

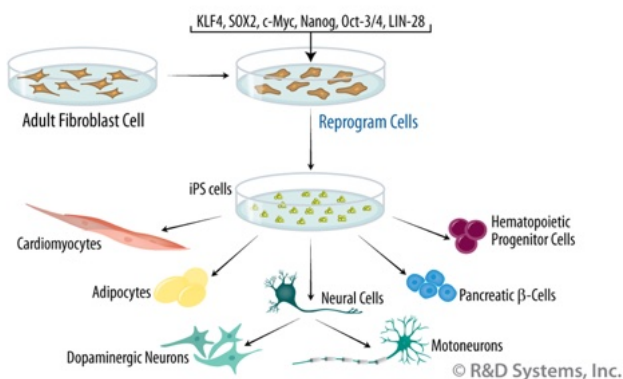


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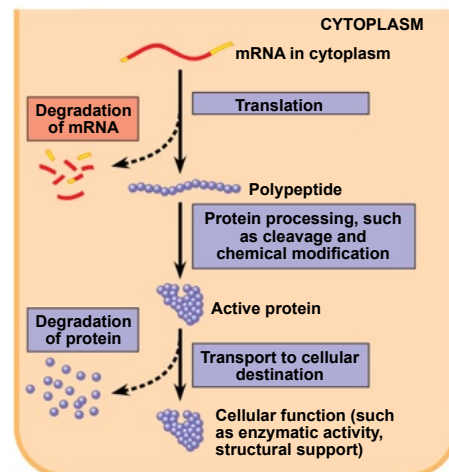
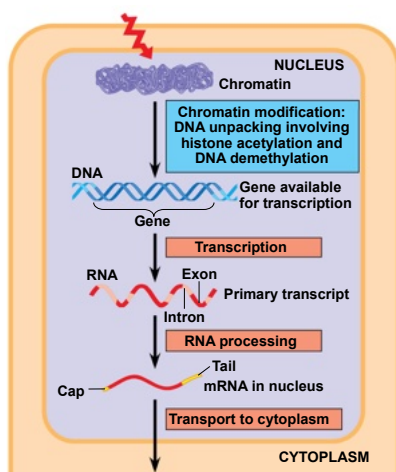
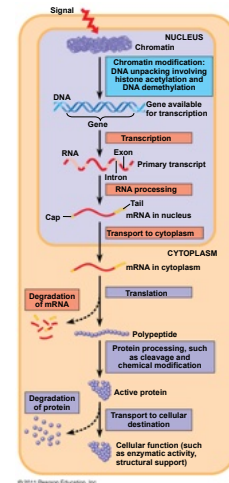
(B)



Generation of stem cells from differentiated tissue (Induced pluripotent stem cells or iPSCs):



Regulation of gene expression is multi-layered and complex!



Gene expression is regulated at multiple levels:

1. **Epigenetic regulation:** chromatin accessibility and architecture, DNA modifications
2. **Transcriptional regulation:** promoters and enhancers, silencers.
3. **Post transcriptional regulation:** alternative splicing, miRNAs
4. **Post-translational regulation:** protein modifications, protein degradation, protein-protein interactions

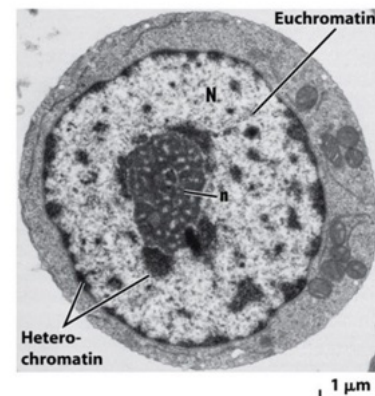
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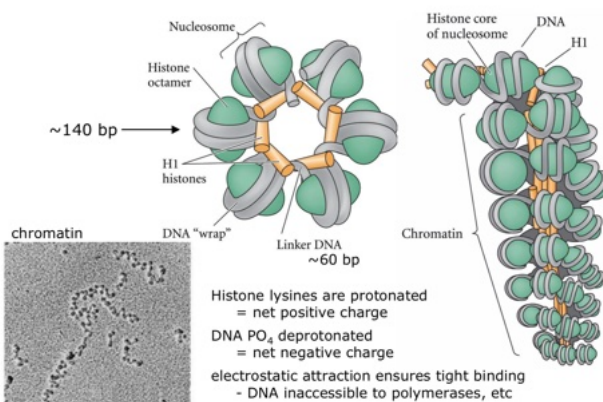
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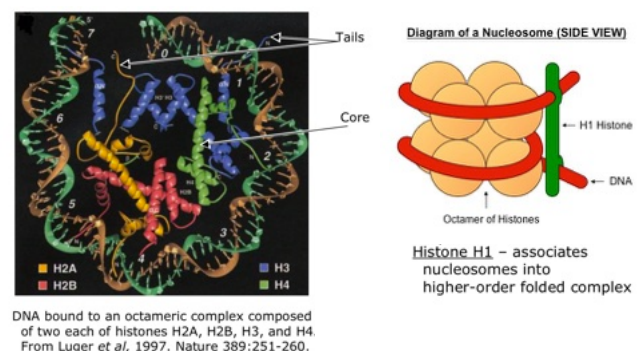
Chromatin organization: heterochromatin vs. euchromatin



DNA is packed with HISTONES to form nucleosomes



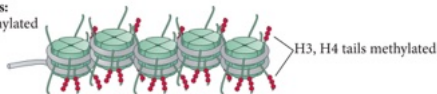
DNA is packed with HISTONES to form nucleosomes



Modification of histone tails determine chromatin condensation

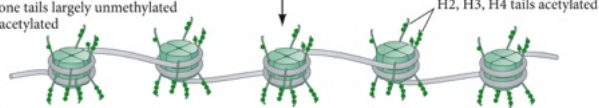
Condensed nucleosomes:

Histone tails largely methylated



Uncondensed nucleosomes:

Histone tails largely unmethylated and acetylated

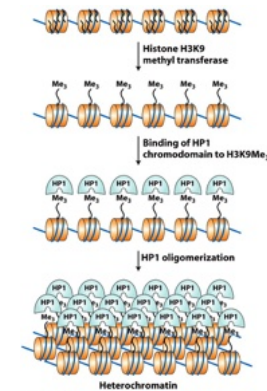


- acetylated histone (**histone acetyltransferase; HAT**) destabilizes = transcription allowed

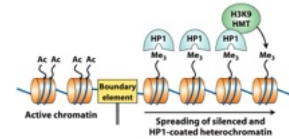
- deacetylated histones (**histone deacetylases; HDAC**) stabilizes = transcription repressed

Histone methylation (CH_3) – transcriptional repression (**histone methyl transferases, HMTs**)

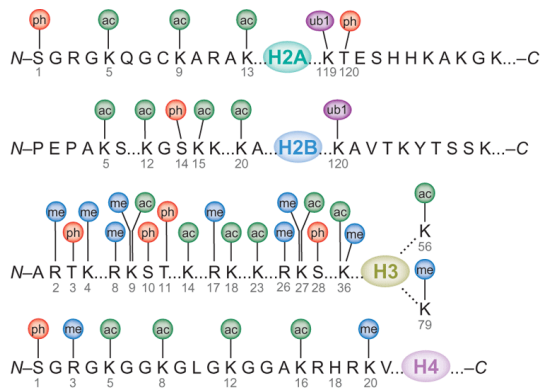
Modification of histone tails determine chromatin condensation



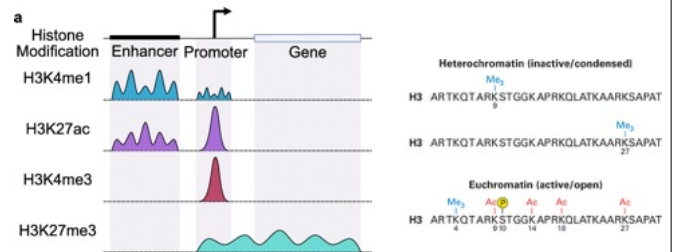
Chromatin condensation into heterochromatin: H3K9Me₃ sites are bound by **heterochromatin protein 1 (HP1)** which oligomerizes resulting in heterochromatin. Recruitment of the H3K9 **histone methyl transferase (HMT)** to HP1 sites promotes heterochromatin spreading by catalyzing H3 methylation.



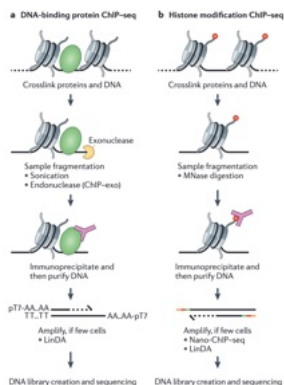
HISTONE tails are modified to give rise to the histone code



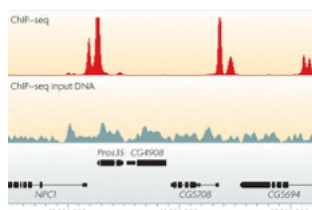
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How are histone marks mapped?

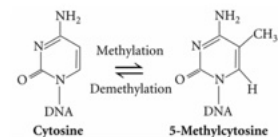


Sequencing results (H3K27AC):



DNA Methylation

DNA methylation stabilizes nucleosomes; prevents transcription

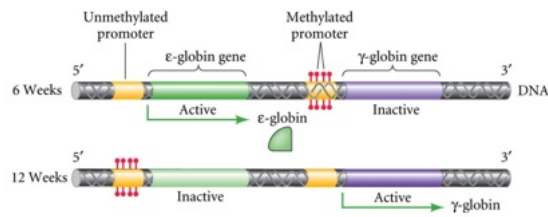


- found in all vertebrates; not *Drosophila*, nematodes, invertebrates
- in adult tissues cytosine methylation occurs only in CpG sequences
 - 1-5% of CpG in adult somatic cells are methylated
- in embryonic tissues and stem cells non-CpGs nucleotides (CpA, CpT) are also methylated
- tumor suppressor genes silenced by methylation in many cancers

Embryonic DNA Methylation

Methylation patterns differ in stage- and tissue-specific manner

- methylation correlates inversely with expression
- methylation pattern changes during development



Methylation patterns are established and maintained throughout cell division by DNA (cytosine-5)-methyltransferases

- humans: DNMT3a, DNMT3b act as *de novo* methyltransferases
- DNMT1 maintains pattern during DNA replication

Gene expression is regulated at multiple levels:

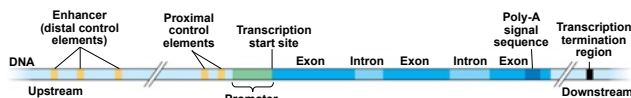
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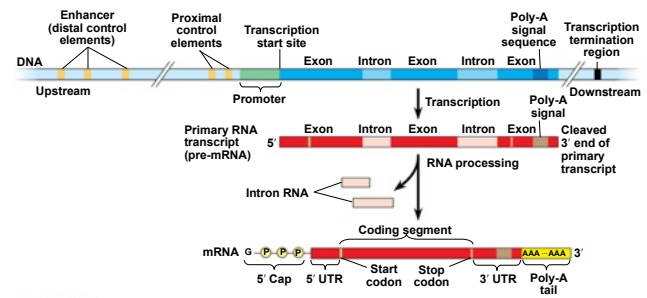
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Organization of a typical eukaryotic gene



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Organization of a typical eukaryotic gene



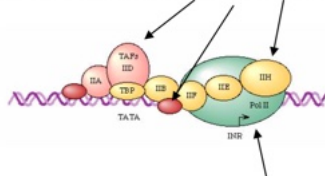
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Promoters

RNA polymerase II binds to the *promoter* region at the TATA box

- however, Pol II cannot initiate transcription alone
- various proteins bind to regulatory sequences upstream and downstream of transcription initiation site

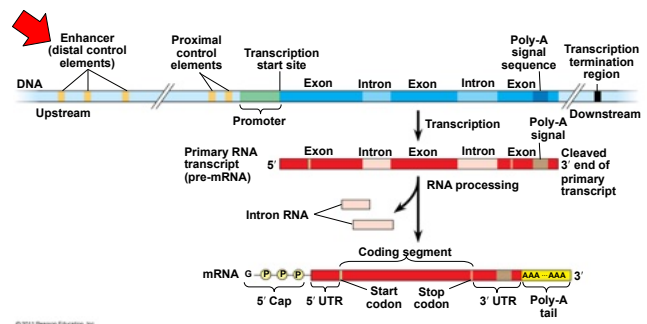
- promoter region binds **Basal Transcription Factors**



- basal transcription factors facilitate Pol II binding and activity

But what controls when and where a promoter is active?

Organization of a typical eukaryotic gene



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Enhancers

Enhancers are *cis-acting regulatory elements*

cis – (same or same side); elements that reside on the same DNA strand;
e.g. DNA sequences
trans – (other side); elements that originate from another DNA strand,
e.g. regulatory proteins

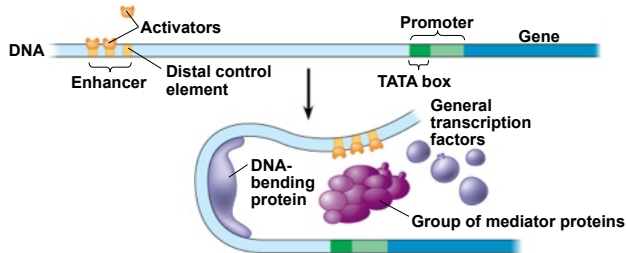
Enhancers: DNA sequences that regulate gene expression by affecting the transcription initiation complex on the promoter.

Function - bind specific regulatory proteins (i.e. *specific Transcription Factors*)
- TFs generally interact with mediator complex proteins – recruit RNA polII

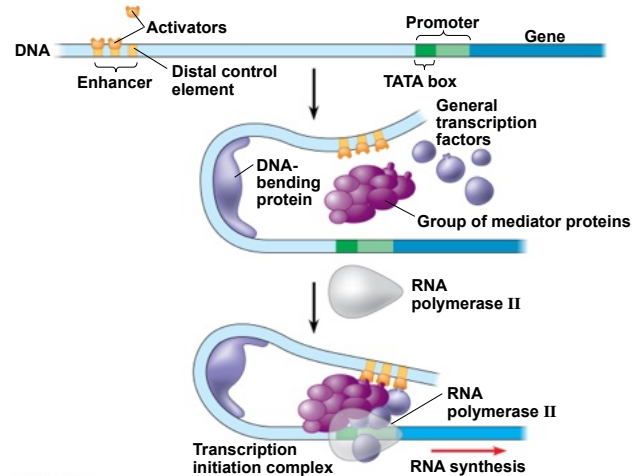
Location - highly variable with respect to the transcribed portion of the gene.
- upstream (5'), downstream (3'), or within transcribed region
- close proximity or as many as 10⁶ bp away



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Enhancers

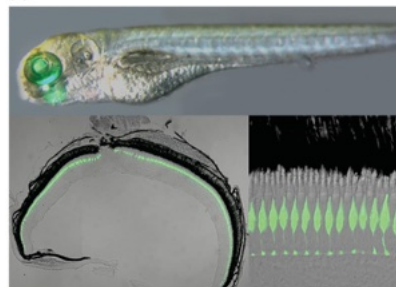
Enhancers differ from promoters:

- 1) need a promoter to work
- 2) can work at a distance
- 3) can work in reverse orientation

Control of gene expression is modular:
a single gene has multiple enhancers

Enhancers control spatial gene expression in the embryo

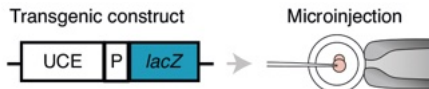
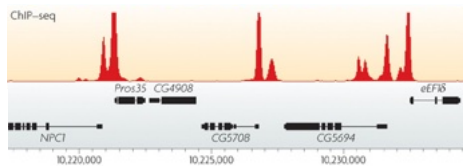
(A)



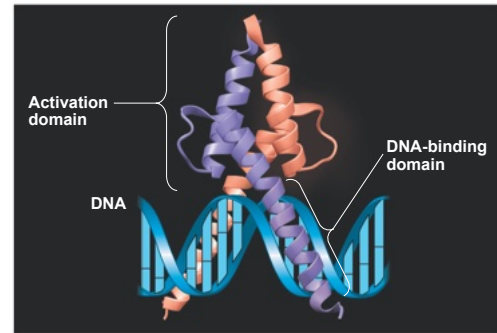
DEVELOPMENTAL BIOLOGY 11e, Figure 3.9
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How do we identify enhancers?



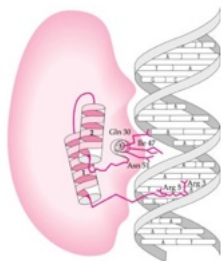
Enhancer activity is regulated by [transcription factors](#)



TRANSCRIPTION FACTOR DOMAINS

Three major domains:

1. DNA-binding
 - recognizes particular DNA sequence

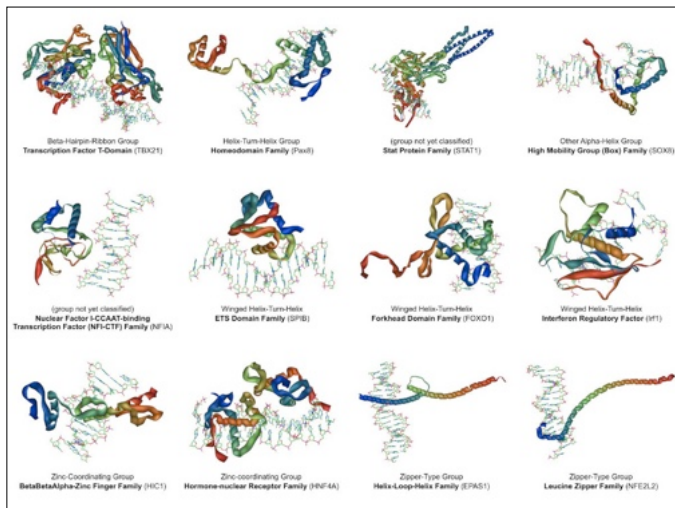


transcription factor - engrailed

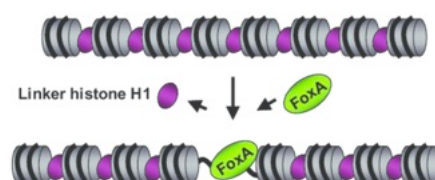
TRANSCRIPTION FACTOR DOMAINS

Three major domains:

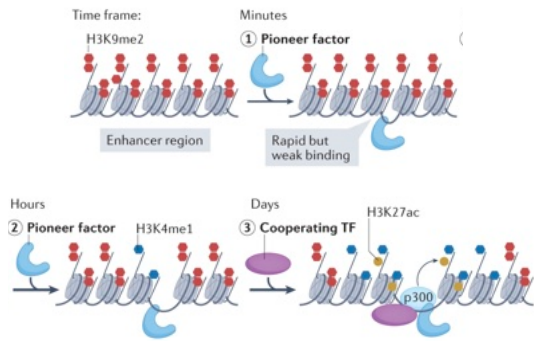
1. DNA-binding
 - recognizes particular DNA sequence
2. Trans-activation
 - activates or represses transcription
 - often involved with transcription coregulator proteins involved in binding RNA polymerase II; e.g. TFIIB, TFIIE
 - often involved with enzymes that modify histones
3. protein-protein interaction domain
 - promotes dimerization
 - allows it to be modulated by TAFs or other transcription factors



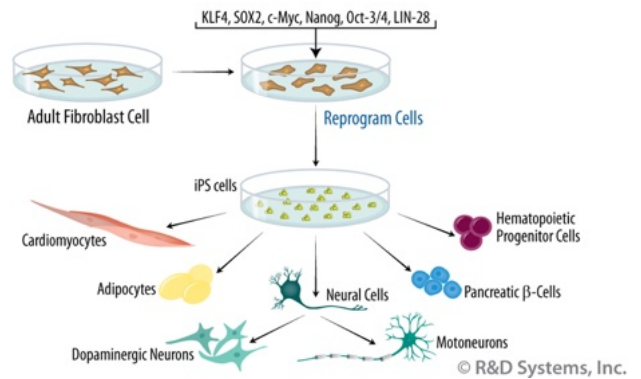
Pioneer transcription factors can open heterochromatin



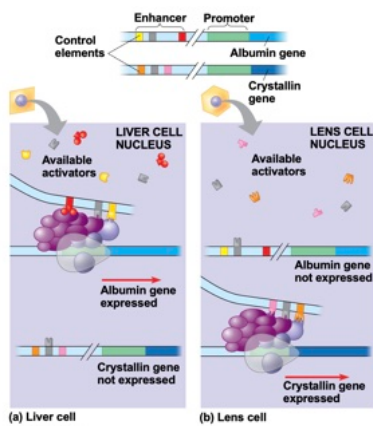
Pioneer transcription factors can open heterochromatin



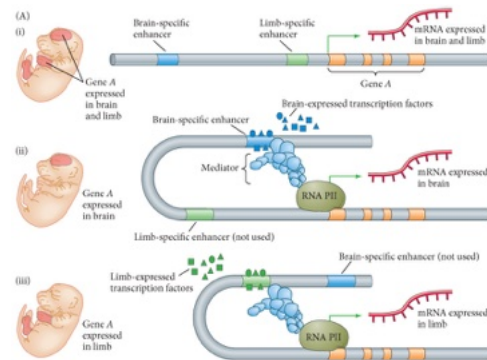
Generation of stem cells from differentiated tissue (Induced pluripotent stem cells or iPSCs):



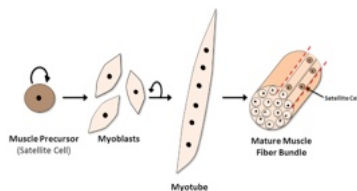
Differential activation of tissue-specific genes



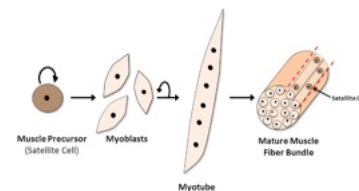
Gene expression is modular



Sequential Regulation of Gene Expression During Cellular Differentiation (an example):



Sequential Regulation of Gene Expression During Cellular Differentiation (an example):



Myoblasts produce muscle-specific proteins and form skeletal muscle cells

MyoD is a regulatory gene that produces proteins that drive commitment to skeletal muscle

The *MyoD* protein is a transcription factor that binds to enhancers of various target genes

